

Question ID: 911383f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$(x - 4)(x + 2)(x - 1) = 0$

What is the product of the solutions to the given equation?

- Answer
- A. 8
 - B. 3
 - C. -3
 - D. -8

Correct Answer: D

Rationale

Choice D is correct. By the zero-product property, if $(x - 4)(x + 2)(x - 1) = 0$, then $x - 4 = 0$, $x + 2 = 0$, or $x - 1 = 0$. Solving each of these equations for x yields $x = 4$, $x = -2$, or $x = 1$. The product of these solutions is $(4)(-2)(1) = -8$.

Choice A is incorrect and may result from sign errors made when solving the given equation. Choice B is incorrect and may result from finding the sum, not the product, of the solutions. Choice C is incorrect and may result from finding the sum, not the product, of the solutions in addition to making sign errors when solving the given equation.